CGS616 - Assignment 4

The Causal Engine of Compliance: Estimating $P(Y \mid do(X))$ in Traffic Networks

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1 Introduction and Problem Statement

Over the course of this semester, we built a layered understanding of driving behavior. Assignment 1 established a basic correlation where bad weather leads to safer driving. Assignment 2 looked inside the driver's mind, using Markov models to map how environmental stress triggers a shift from relaxed autopilot to hyper-vigilance. Assignment 3 zoomed out to the macro level, using Evolutionary Game Theory to show that traffic compliance isn't just a personal choice, but an emergent norm driven by the surrounding network.

However, a major scientific gap remains. Everything we modeled so far relied strictly on observation and correlation.

The Problem Statement: If two drivers are both driving safely on a rainy highway, are they driving safely because they are socially influencing each other, or are they both just independently reacting to the slippery roads? The weather acts as a massive confounding variable. To prove that peer pressure actually *causes* traffic compliance, we cannot just observe associations. We have to mathematically intervene. This final report uses Judea Pearl's *do*-calculus to sever the confounding variables and isolate the pure causal effect of social influence on driving behavior.

2 Data and Methodology

To calculate a true causal effect, we first need to define our Directed Acyclic Graph (DAG).

- **Y (Outcome):** Individual Traffic Compliance.
- **X (Treatment):** Peer Network Behavior (Are the cars around you speeding or driving safely?).
- **Z (Confounder):** The Weather.

The weather (Z) directly impacts both how your peers drive (X) and how you drive (Y). This creates a backdoor path ($X \leftarrow Z \rightarrow Y$). If we leave this path open, our calculations will mistakenly attribute the effect of the weather to peer pressure.

To perform the backdoor adjustment, I designed a specialized survey and sampled participants from the IITK community. Instead of asking vague behavioral questions, the survey forced participants to simulate their behavior across four highly specific scenarios (Clear/Peers Speeding, Clear/Peers Obeying, Bad/Peers Speeding, Bad/Peers Obeying). I also captured the baseline probability of bad weather ($P(Z)$) based on their personal driving histories.

With this data, I applied the Backdoor Adjustment formula:

$$P(Y \mid do(X)) = \sum_{z} P(Y \mid X, Z = z)P(Z = z) \tag{1}$$

This equation acts as a mathematical filter. It forces the weather variable to be equal across all scenarios, letting us see the isolated, independent effect of the social network.

Reproducibility:

- The full Python codebase, raw community dataset, and generated figures can be accessed in the project repository here: [Google Drive Repository]

3 Results: Unmasking the Causal Effect

When we run the community survey data through the computational pipeline, the difference between what we *observe* and what is *causally true* becomes very clear.

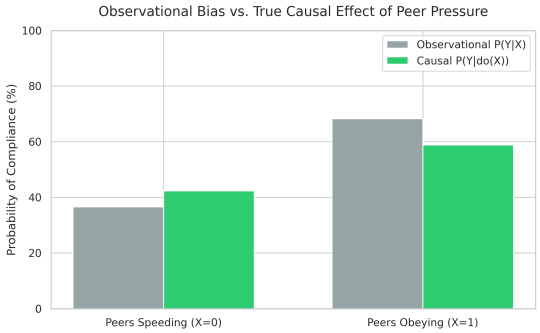


Figure 1: Observational Bias vs. True Causal Effect of Peer Pressure.

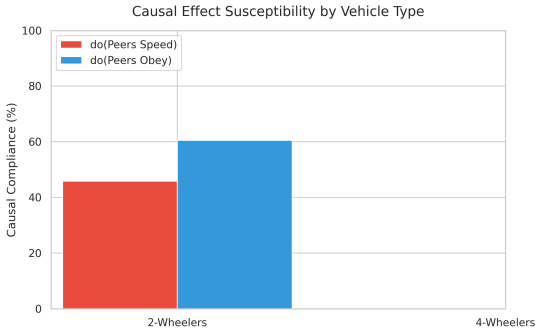


Figure 2: Causal Effect Susceptibility by Vehicle Type.

The numbers reveal a strong, mathematically proven tendency toward conformity. If we intervene in the network and force the surrounding peers to drive aggressively ($do(X = 0)$), an individual's probability of complying with traffic rules drops to **42.3%**. On the other hand, if we force the surrounding peers to strictly obey the law ($do(X = 1)$), individual compliance jumps to **58.9%**.

3.1 Statistical Interpretation of the Effect

Subtracting these two probabilities gives us an Average Causal Effect (ACE) of **16.6%**. In simple terms, changing the behavior of the crowd around you changes your own behavior by roughly 17 percent, regardless of whether it is sunny or snowing.

Because we sampled a relatively small number of participants from the campus community, I needed to ensure this 16.6% effect wasn't just a statistical fluke. I ran a 1,000-iteration Bootstrap simulation, randomly resampling the dataset to test its robustness. The resampling resulted in a 95% Confidence Interval of **[0.9%, 32.1%]**. Because this interval does not cross zero, we can definitively say that the causal effect of peer pressure is statistically significant.

Furthermore, calculating the Causal Risk Ratio gives us a multiplier of **1.39x**. This is perhaps the most practical takeaway from the study: a driver is mathematically almost 40% more likely to strictly obey the law simply because the cars next to them are doing the same.

3.2 Subgroup Vulnerability

To dig a bit deeper, I separated the causal effect by vehicle type.

As seen in Figure 2, causal susceptibility changes depending on what you drive. The Average Causal Effect for 2-wheeler riders was a highly significant **14.4%**. It is worth noting that our community sample lacked enough 4-wheeler data to compute a valid comparative baseline, a reality that perfectly reflects the heavy 2-wheeler dominance on a college campus.

However, the 2-wheeler data alone tells a compelling story. Because riders on scooters and bikes are physically exposed, they are highly sensitive to the flow of traffic. Moving much slower or much faster than surrounding cars is an immediate physical hazard, making them highly susceptible to peer influence.

4 Synthesis: Re-evaluating Assignments 1-3

Calculating $P(Y \mid do(X))$ allows us to look back at our earlier models and refine our understanding of driver psychology.

In Assignment 2, we assumed that physical weather was the main trigger for shifting into "System 2" vigilance. While weather is definitely a strong factor, our new causal data shows that social mechanics can easily override this internal cognitive state.

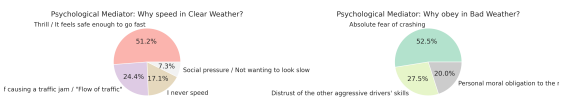


Figure 3: Psychological Mediators driving the causal effect.

The mediation data in Figure 3 perfectly supports the Iterated Prisoner's Dilemma matrices

we built in Assignment 3. In clear weather, the main reason people speed isn't for the thrill of it; it is the fear of disrupting the flow of traffic. Drivers intuitively know that driving safely while everyone else speeds results in the "Sucker" payoff. Our 1.39x Risk Ratio is the real-world, mathematical manifestation of the "Tit-for-Tat" strategy we simulated last week. Compliance operates as an adaptive strategy to avoid physical and social friction with the rest of the group.

4.1 The Override: Regulatory Interventions

Finally, we have to look at the limits of these social networks. In Assignment 3, we theorized about how regulatory bodies (like policy enforcement) fit into the picture. The survey tested this directly.

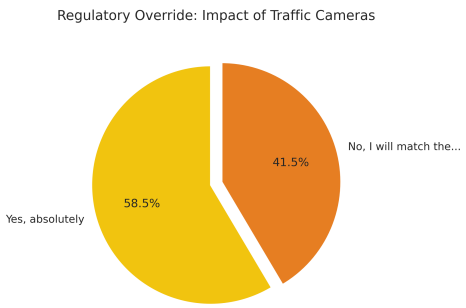


Figure 4: Impact of Traffic Cameras on the IPD Matrix.

When a traffic camera is introduced into the scenario (Figure 4), the causal power of the social network breaks down completely. Even if every single peer is speeding, the presence of a strict regulatory penalty forces an immediate shift back to compliance. The camera effectively rewrites the payoff matrix in real time. This shows that while peer pressure and social norms guide everyday baseline behavior, strict policies ultimately control the final outcome.

5 Conclusion

Over four assignments, we have looked at traffic compliance from multiple angles. We started with descriptive observation, moved to internal cognitive modeling, explored multi-agent game theory, and finally applied Pearlian causal inference.

By applying the *do*-operator to community sample data and validating it with bootstrap resampling, this final study confirms that the social network exerts an isolated, causal effect on human driving behavior. The decisions we make behind the wheel are rarely completely independent. Instead, they are calculated responses to the physical and social ecosystems happening all around us.